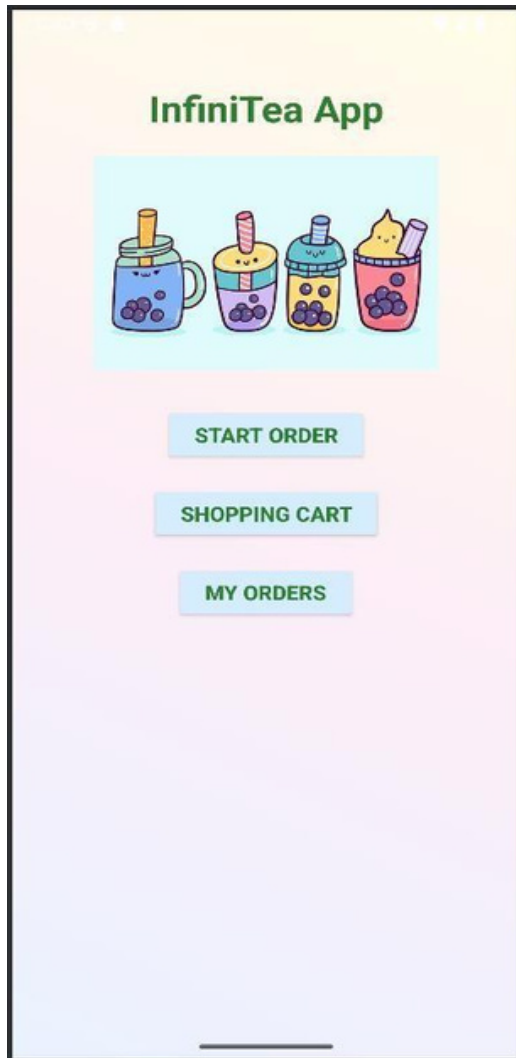


InfiniTea App Summary Report

We made this simple overview of our InfiniTea app project. We built an Android bubble tea ordering app using what we learned from our Software Engineering course. Each screen below shows a different part of the app and what it does.



This is the home screen where users can start an order, view the cart, or check past orders.



This is the menu page with all classic milk teas. The users can pick from the preset drinks. There's also an option to build your own.

← Choose Your Drink

Price:
S: \$5.25
M: \$6.00
L: \$6.75

Black Bubble Milk Tea

A rich black tea with chewy tapioca pearls.
[BUBBLES]

Rose Bubble Milk Tea

Floral rose tea blended with milk and chewy
tapioca pearls.[BUBBLES]

Thai Bubble Milk Tea

Creamy Thai tea with a bold flavor and chewy
tapioca pearls.[BUBBLES]

Jasmine Bubble Milk Tea

Light and fragrant jasmine green tea with chewy
tapioca pearls.[BUBBLES]

Black Bubble Milk Tea

Size
S

Sugar
100%

Ice
100%

- 1 +

ADD TO CART

BUILD YOUR OWN

Once users click any of milk teas from the menu list, then they can customize size, sugar, ice, and quantity.

← Build Your Own

MAKE YOUR DRINK

Available Toppings

BUBBLES

CRYSTAL BOBA

POPPING BOBA

MANGO JELLY

GRASS JELLY

PUDDING

ALOE

Selected Toppings

Total Price
Subtotal: \$0.00
Tax: \$0.00
Total: \$0.00

ADD TO CART

This is the 'Build Your Own' drink page. Users can customize a drink by tapping the "make your drink" button.

12:04

← Build Your Own

MAKE YOUR DRINK

Available Toppings

BUBBLES

Customize Your Drink

Tea Base
ROSE TEA

Size
S

Sugar
100%

Ice
100%

Quantity
- 1 +

CANCEL CONFIRM

ADD TO CART

This pop-up lets users choose tea base, size, sugar, ice, and quantity before adding the drink.

← Build Your Own

MAKE YOUR DRINK

Available Toppings

CRYSTAL BOBA

POPPING BOBA

GRASS JELLY

ALOE

Selected Toppings (3/3)

BUBBLES

PUDDING

MANGO JELLY

Total Price

Subtotal: \$6.25

Tax (8.88%): \$0.55

Total: \$6.80

ADD TO CART

After picking toppings (up to 3), the price updates in real-time.

← Shopping Cart

Customized drink:

• Tea Base: ROSE TEA

• Size: S

• Sugar: 100%

• Ice: 100%

• Toppings: BUBBLES, PUDDING, MANGO JELLY

• Price: \$6.25

Customized drink:

• Tea Base: ROSE TEA

• Size: S

• Sugar: 100%

• Ice: 100%

• Toppings: BUBBLES, PUDDING, MANGO JELLY

• Price: \$6.25

Thai Bubble Milk Tea

• Size: L

• Sugar: 100%

• Ice: 50%

• Toppings: BUBBLES

• Price: \$6.75

Order #:

1

Subtotal:

\$19.25

Tax:

\$1.71

Total:

\$20.96

CLEAR CART

PLACE ORDER

The shopping cart shows drinks, toppings, prices, and the total with tax. Users can clear or place an order. Also, they can single-tap the item (e.g., Thai Bubble Milk Tea) to remove it from the shopping cart

← My Orders

Order Number

Search by Order Number

Order #: 1

• Customized drink: - \$6.25 (S)

• Tea Base: ROSE TEA

• Sugar: 100%

• Ice: 100%

• Toppings: BUBBLES, PUDDING, MANGO JELLY

• Customized drink: - \$6.25 (S)

• Tea Base: ROSE TEA

• Sugar: 100%

• Ice: 100%

• Toppings: BUBBLES, PUDDING, MANGO JELLY

• Thai Bubble Milk Tea - \$6.75 (L)

• Sugar: 100%

• Ice: 50%

• Toppings: BUBBLES

Subtotal: \$19.25

Tax: \$1.71

Total: \$20.96

Date: 2025-05-15

PRINT RECEIPT

REORDER

DELETE ORDER

This is the basic view of a past order, without the checkboxes.

← My Orders

Order Number

1

Order #: 1

• Customized drink: - \$6.25 (S)

• Tea Base: ROSE TEA

• Sugar: 100%

• Ice: 100%

• Toppings: BUBBLES, PUDDING, MANGO JELLY

☒ Add

• Customized drink: - \$6.25 (S)

• Tea Base: ROSE TEA

• Sugar: 100%

• Ice: 100%

• Toppings: BUBBLES, PUDDING, MANGO JELLY

☒ Add

• Thai Bubble Milk Tea - \$6.75 (L)

• Sugar: 100%

• Ice: 50%

• Toppings: BUBBLES

☐ Add

Subtotal: \$19.25

Tax: \$1.71

Total: \$20.96

Date: 2025-05-15

PRINT RECEIPT

REORDER

DELETE ORDER

Once users enter the correct order number to access the order, they can click the check box to select the drink for reordering by tapping the reorder button. This action adds the drinks back into the shopping cart exactly as before.

← My Orders

Order Number

Search by Order Number

Order #: 1

• Customized drink: - \$6.25 (S)

• Tea Base: ROSE TEA

• Sugar: 100%

• Ice: 100%

• Toppings: BUBBLES, PUDDING, MANGO JELLY

• Customized drink: - \$6.25 (S)

• Tea Base: ROSE TEA

• Sugar: 100%

• Ice: 100%

• Toppings: BUBBLES, PUDDING, MANGO JELLY

• Thai Bubble Milk Tea - \$6.75 (L)

• Sugar: 100%

• Ice: 50%

• Toppings: BUBBLES

Subtotal: \$19.25

Tax: \$1.71

Total: \$20.96

Date: 2025-05-15

Order #: 2

• Customized drink: - \$6.25 (S)

• Tea Base: ROSE TEA

• Sugar: 100%

• Ice: 100%

• Toppings: BUBBLES, PUDDING, MANGO JELLY

• Customized drink: - \$6.25 (S)

PRINT RECEIPT

REORDER

DELETE ORDER

There are two past orders on the “My Orders” page. Users can scroll down to see more details. They can also print the receipt or delete any order from the past order list.

Architecture and Design

For this project, we used a lot of ideas from our Software Engineering class. Our app followed the Model-View-Controller (MVC) architecture to ensure ---

- Model classes handle our core data, the necessary tea data. For example, Milk Tea refers to tea name, topping types, sugar and size percentages.
- View components referred to our user interface by a TextView, ListView, RecyclerView.
- Controller elements were the bridge between our data and ui. This was done in Activities.java files like MainActivity or ShoppingCartActivity.

We also applied the Single Responsibility Principle (SRP) by ensuring each class and screen serve one specific purpose. For instance, BuildYourOwnActivity focused only on custom drink creation, and menuRepository was only responsible for accessing and managing menu data.

App Structure and Modularity

We broke the app into different parts like preset drinks (ChooseDrinkActivity), custom drinks (BuildYourOwnActivity), the shopping cart (ShoppingCartActivity), and order history (MyOrdersActivity). This modular approach made development, testing, and debugging much more efficient.

UI Design and User Experience

To ensure our app looked and felt good, we followed Material Design guidelines for the layout, buttons, and overall UI. We also included user feedback and error handling - for example, if someone tries to add more than three toppings, the app stops them and shows a warning message.

Testing and Maintainability

We wrote unit tests and used Espresso tests to ensure everything worked as expected. The code is easy to reuse and extend because we used adapters like MenuAdapter and OrderAdapter, as well as helper classes like OrderFilter and ReorderCartManage, to keep things organized.

Teamwork and Tools

For teamwork, we used Git for version control, Discord to communicate and share feedback, and Google Drive/Docs to plan, share files, and document ideas. Each teammate worked on different parts - UI, backend logic, or testing - and we did code reviews to keep everything running smoothly. It was a great way to turn classroom ideas into a real, working app.